1.

D	y_1	y_2	y_{num}	y_{class}
x_1	1	1	1.25	B
x_2	1	3	7.0	A
x_3	3	2	2.7	C
x_4	3	3	3.2	A
x_5	2	4	5.5	B

$$\underbrace{w = (X^T \cdot X)^{-1} \cdot X^T \cdot z}_{\text{Moore-Penrose solution}}$$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 3 & 3 \\ 1 & 2 & 4 \end{bmatrix} \qquad z = \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix}$$

$$w = \left(\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 3 & 3 \\ 1 & 2 & 4 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} =$$

$$= \left(\begin{bmatrix} 5 & 10 & 13 \\ 10 & 24 & 27 \\ 13 & 27 & 39 \end{bmatrix} \right)^{-1} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} =$$

$$\begin{bmatrix} \frac{23}{11} & \frac{-13}{33} & \frac{-14}{33} \\ \frac{-13}{33} & \frac{26}{99} & \frac{-5}{99} \\ \frac{-14}{33} & \frac{-5}{99} & \frac{20}{99} \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{14}{11} & \frac{14}{33} & \frac{2}{33} & \frac{-4}{11} & \frac{-13}{33} \\ \frac{-2}{11} & \frac{-28}{99} & \frac{29}{99} & \frac{8}{33} & \frac{-7}{99} \\ \frac{-3}{11} & \frac{13}{99} & \frac{-17}{99} & \frac{1}{33} & \frac{28}{29} \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} = \begin{bmatrix} \frac{46}{33} \\ \frac{-1019}{990} \\ \frac{3499}{1980} \end{bmatrix}$$

$$\text{output}(y_1, y_2) = \frac{46}{33} - \frac{1019}{990} \cdot y_1 + \frac{3499}{1980} \cdot y_2$$

2. Ridge regression key difference is in the loss function

$$E(w) = \frac{1}{2}\sum_{i=1}^{N}(z_i - \hat{z}_i)^2 + \frac{\lambda}{2}\|w\|^2$$

$\downarrow$ Moore Penrose solution

$$w = (X^T \cdot X + \lambda \cdot I)^{-1} \cdot X^T \cdot z$$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 3 & 3 \\ 1 & 2 & 4 \end{bmatrix} \quad z = \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} \quad \lambda = 1$$

$$w = \left(\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 3 & 3 \\ 1 & 2 & 4 \end{bmatrix} + \lambda \cdot I \right)^{-1} \cdot X^T \cdot z$$

$$= \left(\begin{bmatrix} 5 & 10 & 13 \\ 10 & 24 & 27 \\ 13 & 27 & 39 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)^{-1} \cdot X^T \cdot z$$

$$= \left(\begin{bmatrix} 6 & 10 & 13 \\ 10 & 25 & 27 \\ 13 & 27 & 40 \end{bmatrix} \right)^{-1} \cdot X^T \cdot z =$$

$$= \begin{bmatrix} \frac{271}{421} & \frac{-49}{421} & \frac{-55}{421} \\ \frac{-49}{421} & \frac{71}{421} & \frac{-32}{421} \\ \frac{-55}{421} & \frac{-32}{421} & \frac{50}{421} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 & 2 \\ 1 & 3 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{167}{421} & \frac{57}{421} & \frac{14}{421} & \frac{-41}{421} & \frac{-47}{421} \\ \frac{-10}{421} & \frac{-74}{421} & \frac{100}{421} & \frac{68}{421} & \frac{-35}{421} \\ \frac{-37}{421} & \frac{63}{421} & \frac{-51}{421} & \frac{-1}{421} & \frac{81}{421} \end{bmatrix} \begin{bmatrix} 1.25 \\ 7.0 \\ 2.7 \\ 3.2 \\ 5.5 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{5117}{8420} \\ \frac{-1177}{2105} \\ \frac{13987}{8420} \end{bmatrix} \quad w_1 = \begin{bmatrix} \frac{46}{33} \\ \frac{-1019}{990} \\ \frac{3499}{1980} \end{bmatrix} \quad w_2 = \begin{bmatrix} \frac{5117}{8420} \\ \frac{-1177}{2105} \\ \frac{13987}{8420} \end{bmatrix}$$

Now we compare the norm of weight vectors. Ridge has a lower norm.

$$\|w_1\| = \frac{436619}{71280} = 6.1254$$

$$\|w_2\| = \frac{121992561}{35441820} = 3.4416$$

Even if we remove the bias Ridge also has higher weight values.

$$\|w_1\| = \frac{3279284}{784080} = 4.18$$

$$\|w_2\| = \frac{217801433}{7086400} = 3.07$$

$b = 1.3934$ simple linear regression

$b = 0.6077$ Ridge regression

Both weights and bias are significantly penalized in Ridge regression.

3. $RMSE = \sqrt{\sum_{i=1}^{n} \frac{(z_i - \hat{z}_i)^2}{n}}$

$x_6 = \begin{bmatrix}1\\2\\2\end{bmatrix}$ $x_7 = \begin{bmatrix}1\\1\\2\end{bmatrix}$ $x_8 = \begin{bmatrix}1\\5\\1\end{bmatrix}$

Test RMSE

Simple Linear regression:

$$w = \begin{bmatrix}\frac{46}{33}\\ \frac{-1019}{990}\\ \frac{3499}{1980}\end{bmatrix} \quad \hat{z}_6 = \begin{bmatrix}1\\2\\2\end{bmatrix} \cdot \begin{bmatrix}\frac{46}{33}\\ \frac{-1019}{990}\\ \frac{3499}{1980}\end{bmatrix} = \frac{947}{330} = 2.8697$$

$$\hat{z}_7 = \begin{bmatrix}1\\1\\2\end{bmatrix} \cdot \begin{bmatrix}\frac{46}{33}\\ \frac{-1019}{990}\\ \frac{3499}{1980}\end{bmatrix} = \frac{386}{99} = 3.8990$$

$$\hat{z}_8 = \begin{bmatrix}1\\5\\1\end{bmatrix} \cdot \begin{bmatrix}\frac{46}{33}\\ \frac{-1019}{990}\\ \frac{3499}{1980}\end{bmatrix} = \frac{-3931}{1980} = -1.9854$$

$$RMSE = \sqrt{\frac{(0.7-2.8697)^2 + (1.1-3.8990)^2 + (2.2-(-1.9854))^2}{3}}$$

$$= \sqrt{\frac{4.7076 + 7.83441 + 17.5116}{3}} = 3.1654$$

Ridge Regression

$$w = \begin{bmatrix}\frac{5117}{8420}\\ \frac{-1177}{2105}\\ \frac{13987}{8420}\end{bmatrix} \quad \hat{z}_6 = \begin{bmatrix}1\\2\\2\end{bmatrix} \cdot w = \frac{4735}{1684} = 2.8118$$

$$\hat{z}_7 = \begin{bmatrix}1\\1\\2\end{bmatrix} \cdot w = \frac{28383}{8420} = 3.3709$$

$$\hat{z}_8 = \begin{bmatrix}1\\5\\1\end{bmatrix} \cdot w = \frac{-1109}{2105} = -0.5268$$

$$\sqrt{\frac{(0.7-2.8118)^2 + (1.1-3.3709)^2 + (2.2+0.5268)^2}{3}} =$$

$$= \sqrt{\frac{4.4597}{3} + \frac{5.1570}{3} + \frac{7.435438241}{3}} =$$

$$= \sqrt{5.6840} = 2.384$$

Train RMSE

Ridge:

$$w = \begin{bmatrix}\frac{5117}{8420}\\ \frac{-1177}{2105}\\ \frac{13987}{8420}\end{bmatrix}$$

$$\hat{y}_1 = w \cdot \begin{bmatrix}1\\1\\1\end{bmatrix} = 1.7097$$

$$\hat{y}_2 = w \cdot \begin{bmatrix}1\\1\\3\end{bmatrix} = 5.0321$$

$$\hat{y}_3 = w \cdot \begin{bmatrix}1\\3\\2\end{bmatrix} = 2.2526$$

$$\hat{y}_4 = w \cdot \begin{bmatrix}1\\3\\3\end{bmatrix} = 3.9138$$

$$\hat{y}_5 = w \cdot \begin{bmatrix}1\\2\\4\end{bmatrix} = 6.1341$$

$$RMSE =$$

$$\sqrt{\frac{(1.7097-1.25)^2 + (5.0321-7.0)^2 + (2.2526-2.7)^2 + (3.9138-3.2)^2 + (6.1341-5.5)^2}{5}} =$$

$$= 1.0194$$

= Simple Linear regression:

$$w = \begin{bmatrix}\frac{46}{33}\\ \frac{-1019}{990}\\ \frac{3499}{1980}\end{bmatrix} \quad \hat{y}_1 = w \cdot \begin{bmatrix}1\\1\\1\end{bmatrix} = 2.1318$$

$$\hat{y}_2 = w \cdot \begin{bmatrix}1\\1\\3\end{bmatrix} = 5.6662$$

$$\hat{y}_3 = w \cdot \begin{bmatrix}1\\3\\2\end{bmatrix} = 1.8404 \quad \hat{y}_4 = w \cdot \begin{bmatrix}1\\3\\3\end{bmatrix} = 3.6076$$

$$\hat{y}_5 = w \cdot \begin{bmatrix}1\\2\\4\end{bmatrix} = 6.4040$$

$$\sqrt{\frac{(2.1318-1.25)^2 + (5.6662-7)^2 + (1.8404-2.7)^2 + (3.6076-3.2)^2 + (6.4040-5.5)^2}{5}}$$

$$= 0.9251$$

	Simple	Ridge
Train	0.9251	1.0194
Test	3.1654	2.384

The results were as expected. Ridge regression had worse performance on training data, but could generalize well. It successfully prevented overfitting. The simple linear regression, despite the better performance in training data, lacked the ability to generalize.

4. $W^1 = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} \quad b^1 = \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix}$

$W_2 = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$x_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad W^1 x_n + b^1 = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} =$

$= \begin{bmatrix} 0.2 \\ 0.3 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.3 \\ 0.4 \end{bmatrix} \rightarrow z^1$

$W^2 z^1 + b^2 = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.3 \\ 0.4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} =$

$= \begin{bmatrix} \frac{13}{10} \\ \frac{13}{10} \\ \frac{13}{10} \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{23}{10} \\ \frac{23}{10} \\ \frac{23}{10} \\ 2 \end{bmatrix}$

$\sum_{i=1}^{N} e^{z_i} = e^{\frac{23}{10}} + e^{\frac{23}{10}} + e^2 = 32.24297$

$\frac{e^{\frac{23}{10}}}{32.24297} = 0.4615 \qquad \hat{a} = 1$

$\frac{e^{\frac{23}{10}}}{32.24297} = 0.3093 \qquad \begin{bmatrix} 0.4615 \\ 0.3093 \\ 0.2292 \end{bmatrix}$

$\frac{e^2}{32.24297} = 0.2292$

Our targets for x_1 are: $a_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

To perform a stochastic gradient descent, we have to optimize the following function:

$-\sum_{i=1}^{N} \sum_{t=1}^{|C|} t_i \log(z_i^{out}) =$

$= -\left(t_2 \log z_2^{out}\right) = -\log z_2^{out}$

$t_2 = 1$

$-\log(\text{softmax}(x_2^{out}))$

Last layer:

$\frac{\partial E}{\partial W^{(2)}} = \underbrace{\left(\frac{\partial E}{\partial z^{(2)}} \circ \frac{\partial x^{(2)}}{\partial z^{(2)}}\right)}_{\delta^{(2)}} \underbrace{\left(\frac{\partial z^{(2)}}{\partial W^{(2)}}\right)^T}_{x^{(2)}} =$

$= \delta^{(2)} (x^{(2)})^T$

$\delta^{(2)} = x^{(2)} - t = \begin{bmatrix} 0.4615 \\ 0.3093 \\ 0.2292 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.4615 \\ -0.6907 \\ 0.2292 \end{bmatrix}$

$\frac{\partial E}{\partial W^{(2)}} = \begin{bmatrix} 0.4615 \\ -0.6907 \\ 0.2292 \end{bmatrix} \begin{bmatrix} 0.3 & 0.3 & 0.4 \end{bmatrix} =$

$= \begin{bmatrix} 0.1385 & 0.1385 & 0.1846 \\ -0.2072 & -0.2072 & -0.2763 \\ 0.0688 & 0.0688 & 0.0917 \end{bmatrix}$

$W^{(2)}_{new} = W^{(2)}_{old} - \mu \frac{\partial E}{\partial W^{(2)}} =$

$= \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} - 0.1 \begin{bmatrix} 0.1385 & 0.1385 & 0.1846 \\ -0.2072 & -0.2072 & -0.2763 \\ 0.0688 & 0.0688 & 0.0917 \end{bmatrix}$

$= \begin{bmatrix} 0.9862 & 1.9862 & 1.9815 \\ 1.0207 & 2.0207 & 1.0276 \\ 0.9931 & 0.9931 & 0.9908 \end{bmatrix}$

Remaining layer

$\delta^{(1)} = \left(\frac{\partial z^{(1+1)}}{\partial x^{(1)}}\right)^T \cdot \delta^{(1+1)} \circ \frac{\partial x^{(1)}}{\partial z^{(1)}}$

$= W^{(1+1)T} \cdot \delta^{(1+1)} \circ \frac{\partial x^{(1)}}{\partial z^{(1)}} = W^{(1+1)T} \delta^{(1+1)}$

(no activation function)

$\delta^{(1)} = W^{(2)T} \cdot \delta^2 = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.4615 \\ -0.6907 \\ 0.2292 \end{bmatrix}$

$= \begin{bmatrix} -0.4615 \\ -0.6907 \\ 0 \end{bmatrix}$

$\frac{\partial E}{\partial W^{(1)}} = \delta^{(1)} \cdot (x^{(1-1)})^T$

$\frac{\partial E}{\partial W^{(1)}} = \delta^{(1)} \cdot (x^{(0)})^T =$

$= \begin{bmatrix} -0.4615 \\ -0.6907 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} -0.4615 & -0.4615 \\ -0.6907 & -0.6907 \\ 0 & 0 \end{bmatrix}$

$W^{(1)}_{new} = W^{(1)}_{old} - \mu \frac{\partial E}{\partial W^{(1)}} =$

$= \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} - 0.1 \begin{bmatrix} -0.4615 & -0.4615 \\ -0.6907 & -0.6907 \\ 0 & 0 \end{bmatrix} =$

$= \begin{bmatrix} 0.1462 & 0.1462 \\ 0.1691 & 0.2691 \\ 0.2 & 0.1 \end{bmatrix}$

Bias update:

$\frac{\partial E}{\partial b^{(2)}} = \delta^{(2)} \cdot \frac{\partial z^{(2)}}{\partial b^{(2)}} = \delta^{(2)}$

$b^{(2)}_{new} = b^{(2)}_{old} - \mu \frac{\partial E}{\partial b^{(2)}} = b^{(2)}_{old} - \mu \delta^{(2)}$

$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 0.4615 \\ -0.6907 \\ 0.2292 \end{bmatrix} = \begin{bmatrix} 0.9539 \\ 0.1691 \\ 0.9771 \end{bmatrix}$

$b^{(1)}_{new} = b^{(1)}_{old} - \mu \frac{\partial E}{\partial b^{(1)}} = b^{(1)}_{old} - \mu \delta^{(1)}$

$= \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} - 0.1 \begin{bmatrix} -0.4615 \\ -0.6907 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1462 \\ 0.0691 \\ 0.1 \end{bmatrix}$

Summary of updates in ex. 4

$$W^{[1]} = \begin{bmatrix} 0.1462 & 0.1462 \\ 0.1691 & 0.2691 \\ 0.2 & 0.1 \end{bmatrix}$$

$$b^{[1]} = \begin{bmatrix} 0.1462 \\ 0.0691 \\ 0.1 \end{bmatrix} \qquad b^{[2]} = \begin{bmatrix} 0.9539 \\ 0.9691 \\ 0.9771 \end{bmatrix}$$

$$W^{[2]} = \begin{bmatrix} 0.9862 & 1.9862 & 1.9815 \\ 1.0207 & 2.0207 & 1.0276 \\ 0.9931 & 0.9931 & 0.9908 \end{bmatrix}$$